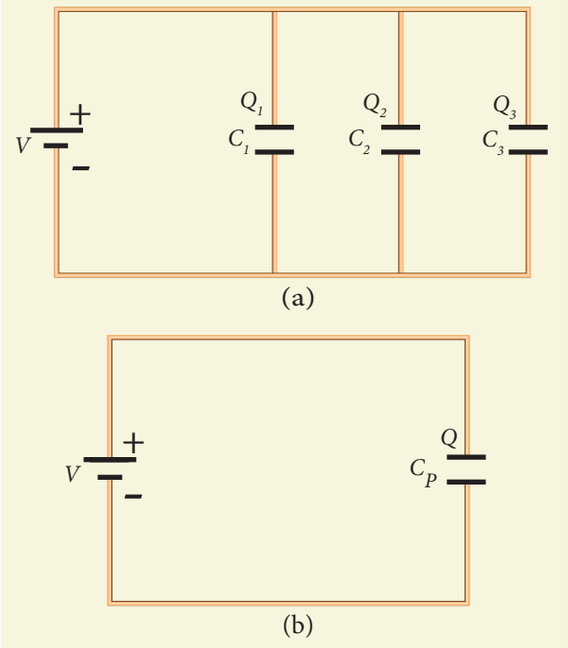


Figure 1.59 (a) capacitors in parallel (b) equivalent capacitance with the same total charge

the charge stored in each capacitor is not the same. Let the charge stored in the three capacitors be Q_1, Q_2, and Q_3 respectively. According to the law of conservation of total charge, the sum of these three charges is equal to the charge Q transferred by the battery,

$$Q = Q_1 + Q_2 + Q_3 \qquad (1.106)$$

Since $Q = CV$, we have

$$Q = C_1V + C_2V + C_3V \qquad (1.107)$$

If these three capacitors are considered to form a single equivalent capacitance C_P which stores the total charge Q as shown in the Figure 1.59(b), then we can write $Q = C_PV$. Substituting this in equation (1.107), we get

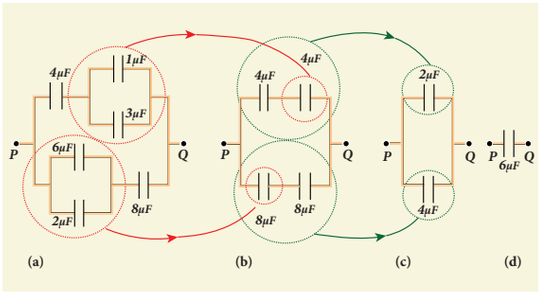
$$C_PV = C_1V + C_2V + C_3V$$

$$C_P = C_1 + C_2 + C_3 \qquad (1.108)$$

Thus, the equivalent capacitance of capacitors connected in parallel is equal to the sum of the individual capacitances. The equivalent capacitance C_P in a parallel connection is always greater than the largest individual capacitance. In a parallel connection, it is equivalent as area of each capacitance adds to give more effective area such that total capacitance increases.

EXAMPLE 1.22

Find the equivalent capacitance between P and Q for the configuration shown below in the figure (a).



Solution

The capacitors 1 μF and 3 μF are connected in parallel and 6 μF and 2 μF are also separately connected in parallel. So these parallel combinations reduced to equivalent single capacitances in their respective positions, as shown in the figure (b).

$$C_{eq} = 1 + 3 = 4\ \mu F$$

$$C_{eq} = 6 + 2 = 8\ \mu F$$

From the figure (b), we infer that the two 4 μF capacitors are connected in series and the two 8 μF capacitors are connected in series. By using formula for the series, we can reduce to their equivalent capacitances as shown in figure (c).

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